MLA0202-Fundamentals Of Machine Learning

Analytical Problem Solving

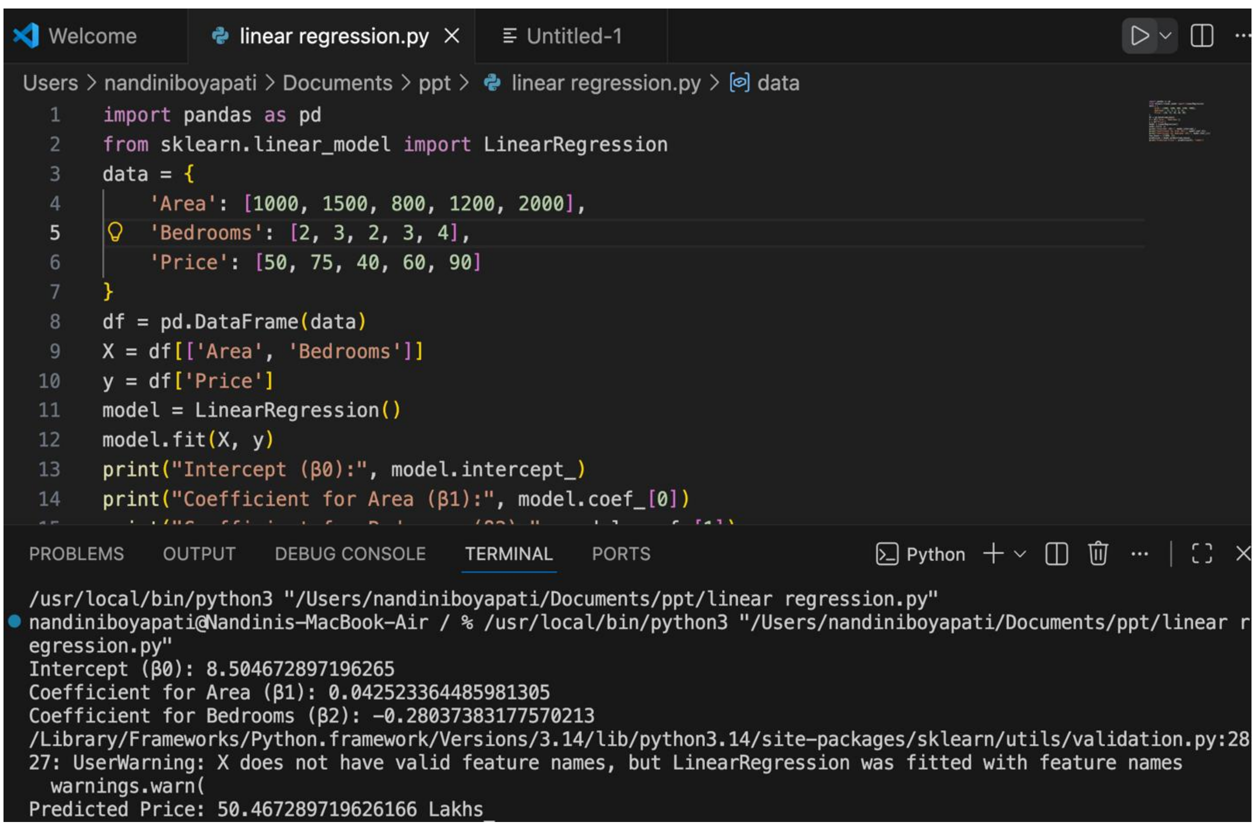
Question 1: Multiple Linear Regression

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
data = {
    'Area': [1000, 1500, 800, 1200, 2000],
    'Bedrooms': [2, 3, 2, 3, 4],
    'Price': [50, 75, 40, 60, 90]
}
df = pd.DataFrame(data)
X = df[['Area', 'Bedrooms']]
y = df['Price']
model = LinearRegression()
model.fit(X, y)
print("Intercept (β0):", model.intercept_)
```

```
print("Coefficient for Area (β1):", model.coef_[0])
print("Coefficient for Bedrooms (β2):", model.coef_[1])
new_house = [[1000, 2]]
prediction = model.predict(new_house)
print("Predicted Price:", prediction[0], "Lakhs")
```

Output:

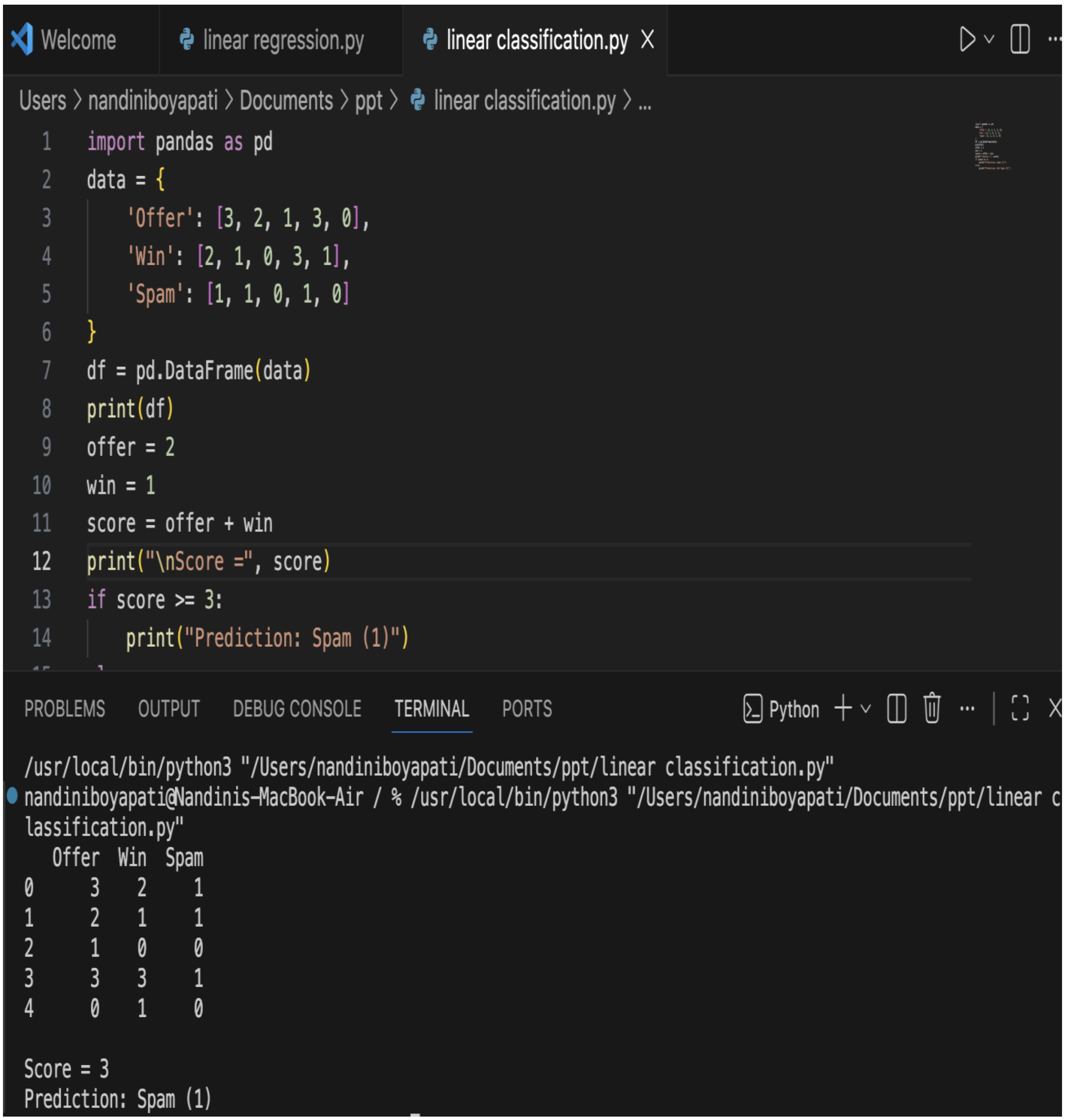


Question 2: Linear Classification

Code:

```
import pandas as pd
data = {
    'Offer': [3, 2, 1, 3, 0],
    'Win': [2, 1, 0, 3, 1],
    'Spam': [1, 1, 0, 1, 0]
}
df = pd.DataFrame(data)
print(df)
offer = 2
win = 1
score = offer + win
print("\nScore =", score)
if score >= 3:
    print("Prediction: Spam (1)")
else:
    print("Prediction: Not Spam (0)")
```

Output:


Welcome
linear regression.py
linear classification.py
Users > nandiniboyapati > Documents > ppt > linear classification.py > ...
import pandas as pd
data = {
'Offer': [3, 2, 1, 3, 0],
'Win': [2, 1, 0, 3, 1],
'Spam': [1, 1, 0, 1, 0]
}
df = pd.DataFrame(data)
print(df)
offer = 2
win = 1
score = offer + win
print("\nScore =", score)
if score >= 3:
print("Prediction: Spam (1)")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
/usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/linear classification.py"
nandiniboyapati@Nandinis-MacBook-Air / % /usr/local/bin/python3 "/Users/nandiniboyapati/Documents/ppt/linear classification.py"
Offer Win Spam
0 3 2 1
1 2 1 1
2 1 0 0
3 3 3 1
4 0 1 0
Score = 3
Prediction: Spam (1)

Question 3 : Naive Bayes Clssification

Code:

```
]import pandas as pd
from sklearn.naive_bayes import BernoulliNB
data = {
    'Fever': [1, 1, 0, 1, 0],
    'Headache': [1, 0, 1, 1, 0],
    'Disease': [1, 1, 0, 1, 0]
}
df = pd.DataFrame(data)
X = df[['Fever', 'Headache']]
y = df['Disease']
model = BernoulliNB()
model.fit(X, y)
test = [[1, 0]]
prediction = model.predict(test)
probability = model.predict_proba(test)
print("Prediction:", "Yes" if prediction[0] == 1 else "No")
```

```
print("Probability:", probability)
```

Output :